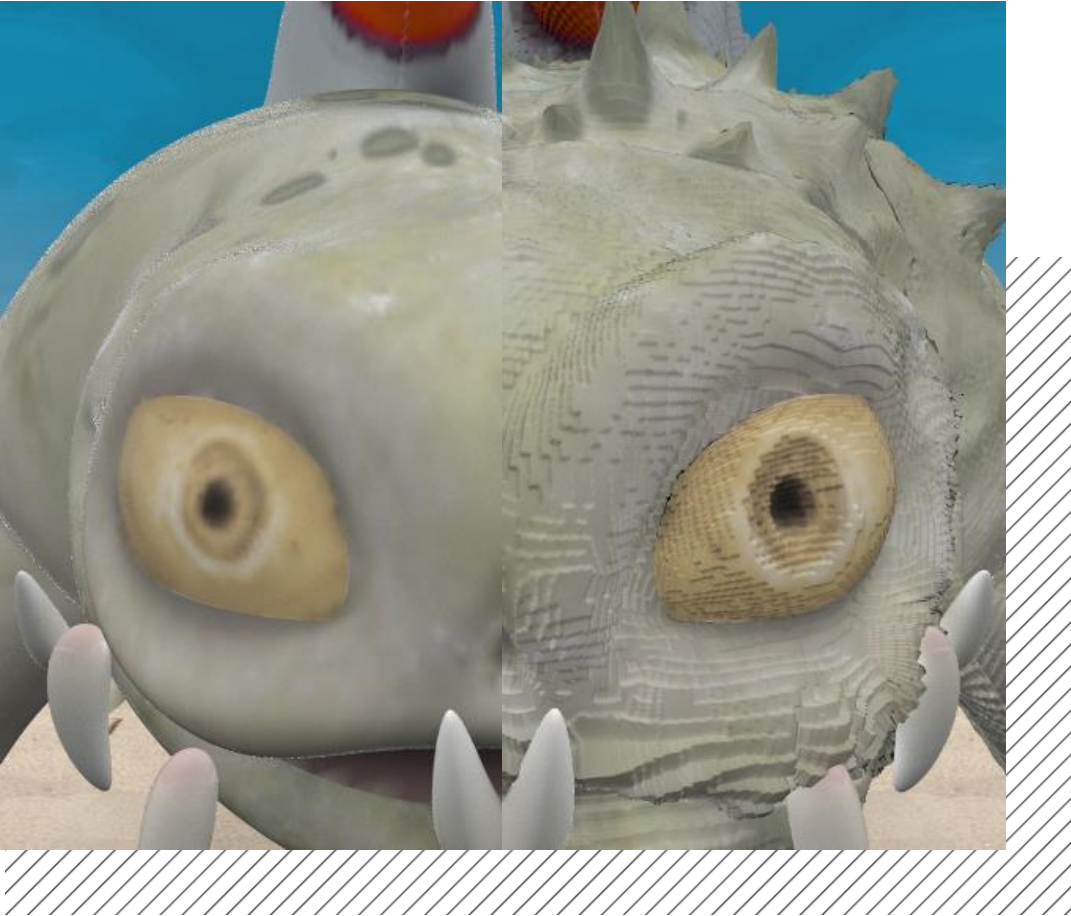




**Ray Tracing of Compressed Subdivision
Surfaces with Displacement on the GPU
using OptiX**

Lukas Wolf

Ray Tracing of Compressed Subdivision Surfaces

| Agenda

1 Ray Tracing of Subdivision Surfaces

2 Data Structures

3 Compression Techniques

4 OptiX Implementation

5 Evaluation

6 Future Work

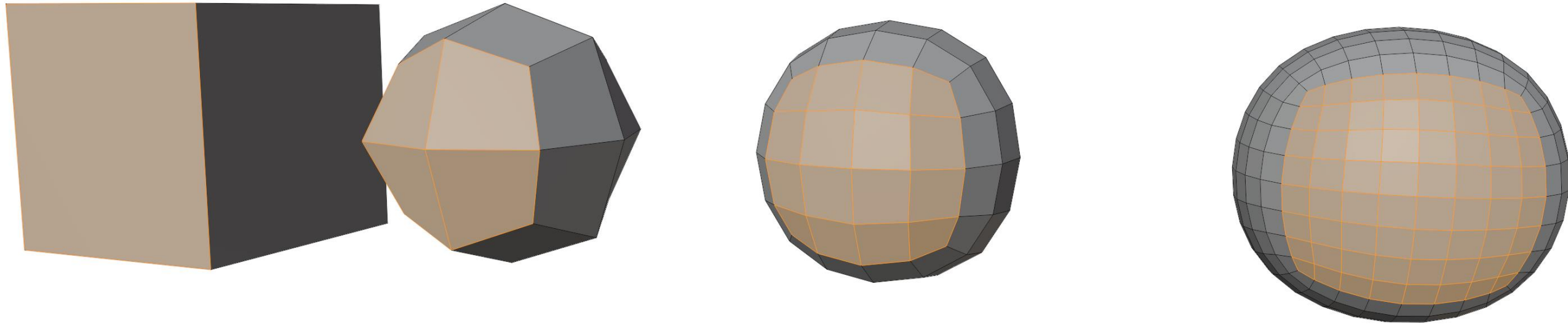
1 Ray Tracing of Subdivision Surfaces

Ray Tracing



1 Ray Tracing of Subdivision Surfaces

| Catmull-Clark Subdivision Surfaces



1 Ray Tracing of Subdivision Surfaces

| Displacement Mapping – Base Mesh



1 Ray Tracing of Subdivision Surfaces

| Displacement Mapping – Subdivided Mesh



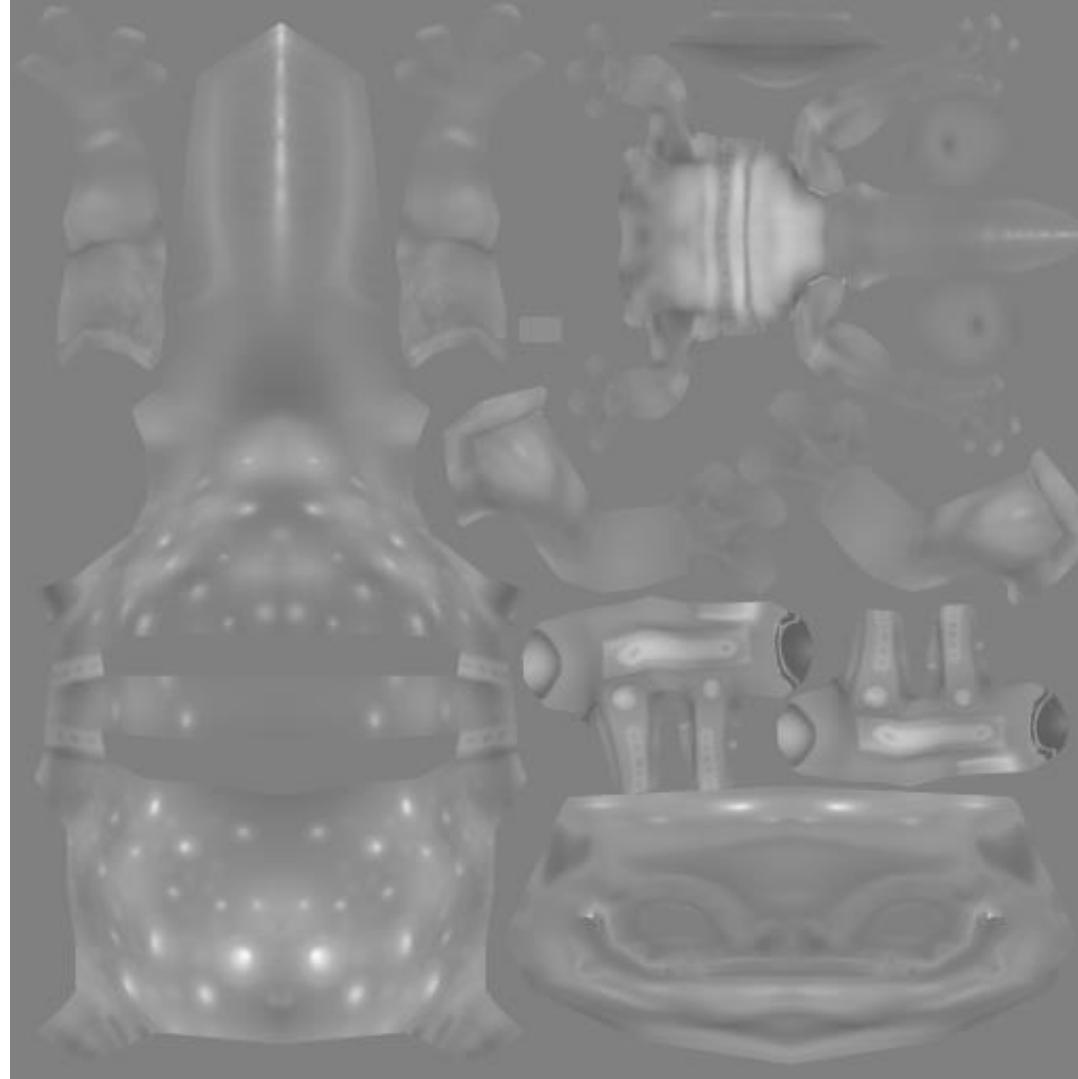
1 Ray Tracing of Subdivision Surfaces

| Displacement Mapping – Subdivided and Displaced Mesh



1 Ray Tracing of Subdivision Surfaces

| Displacement Map



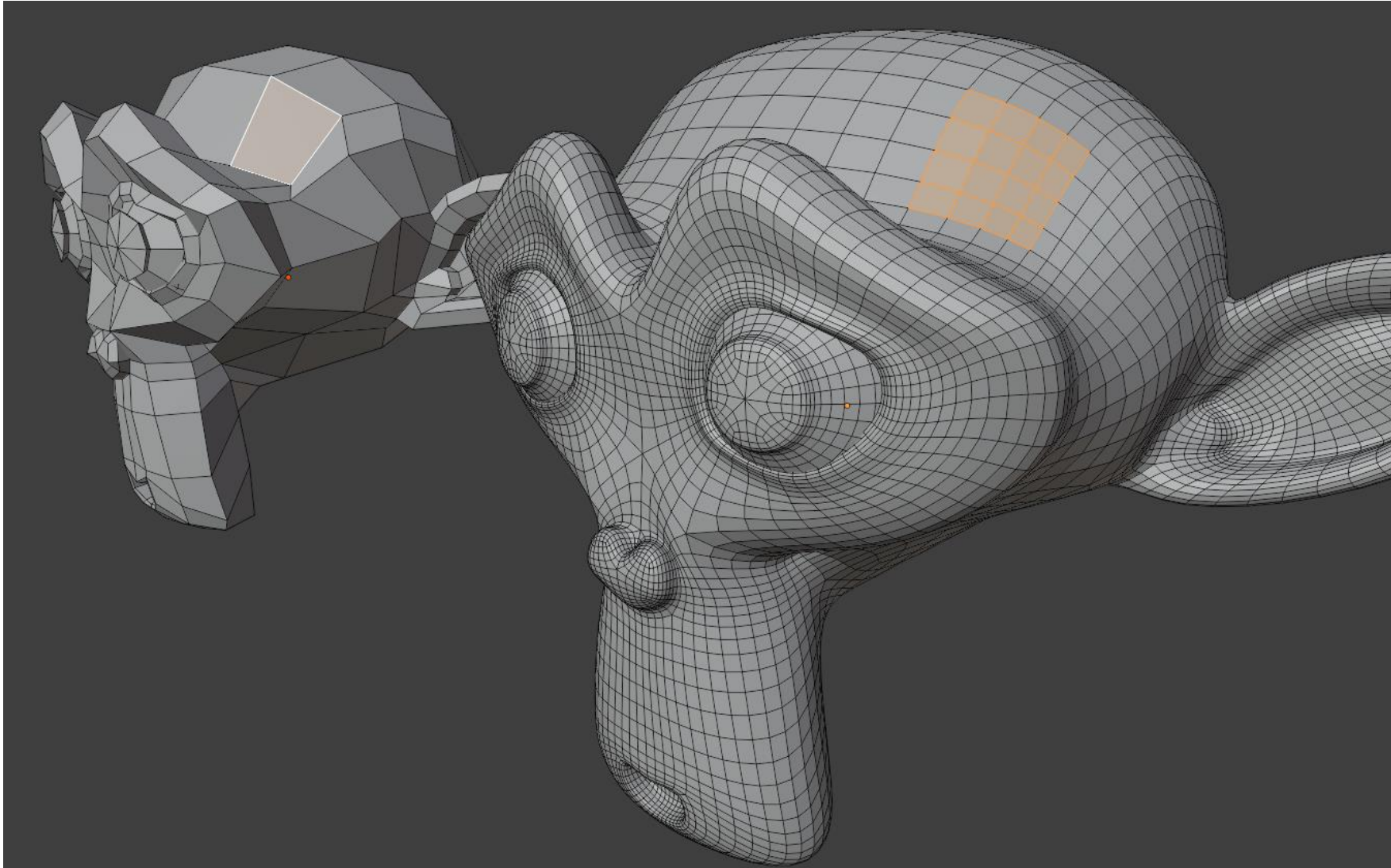
1 Ray Tracing of Subdivision Surfaces

Subdivision Comparison



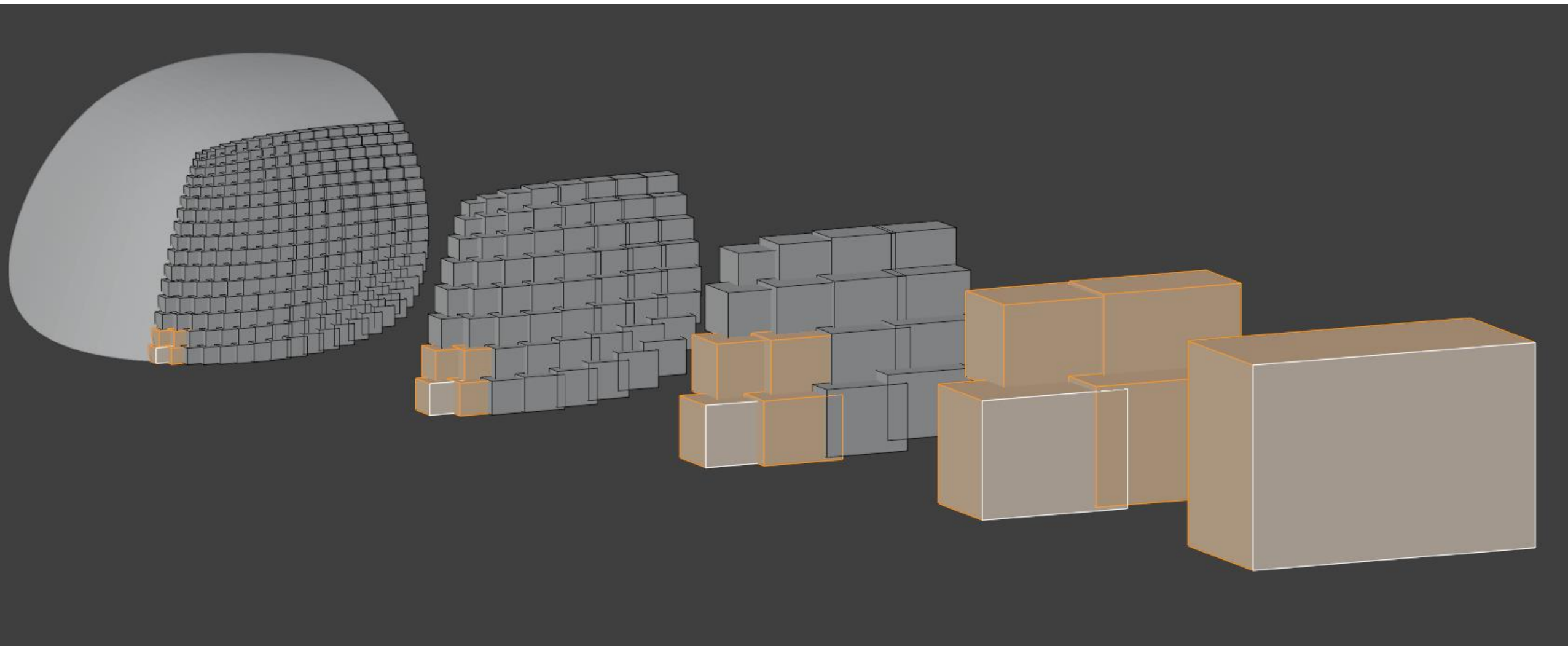
2 Data Structures

Patches and Grid-Layout



2 Data Structures

| Bounding Volume Hierarchy

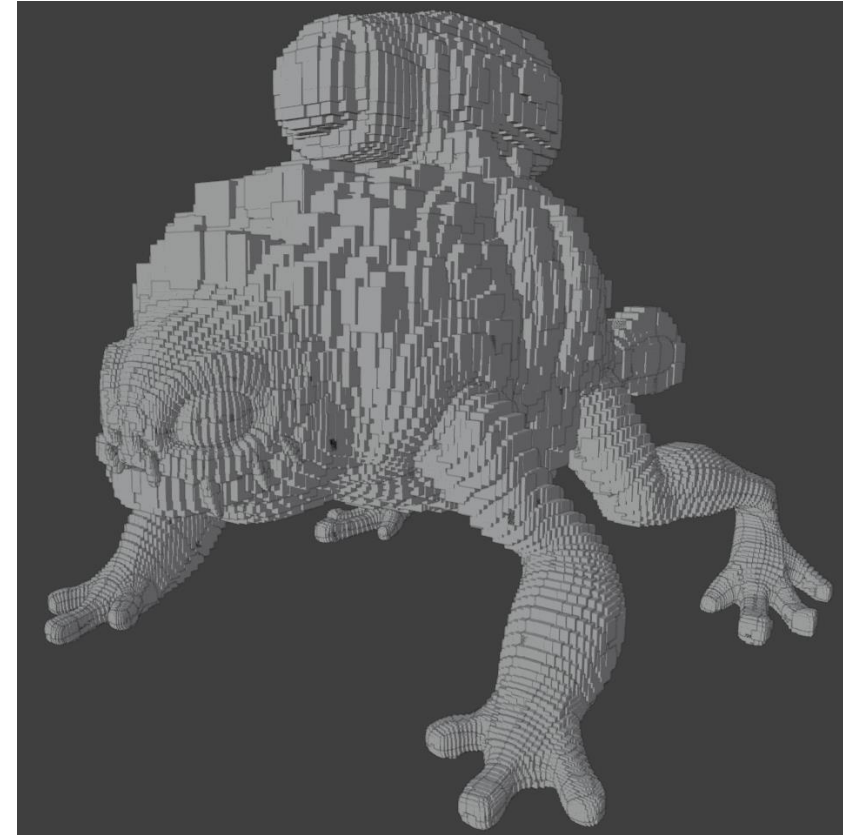


3 Compression Techniques [SLM*16]

Idea



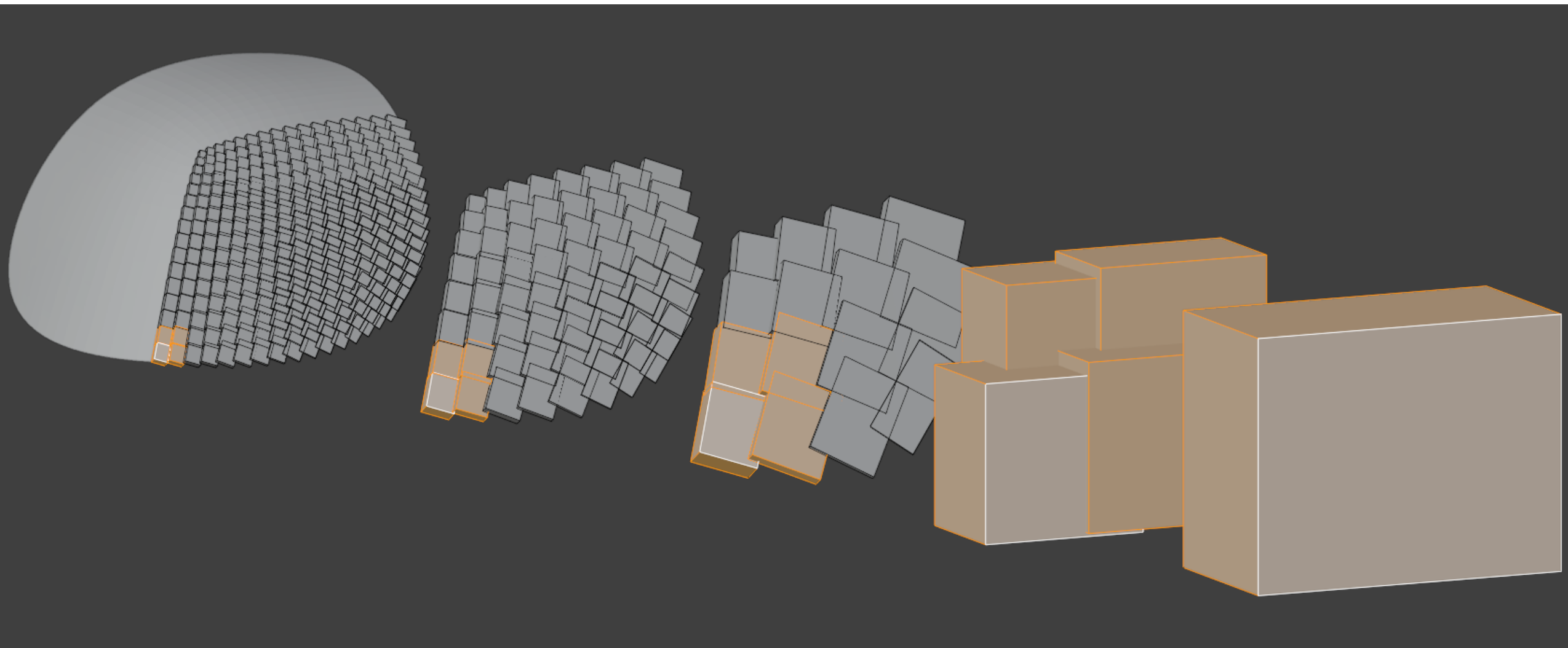
Subdivided Model



Box Approximated Model

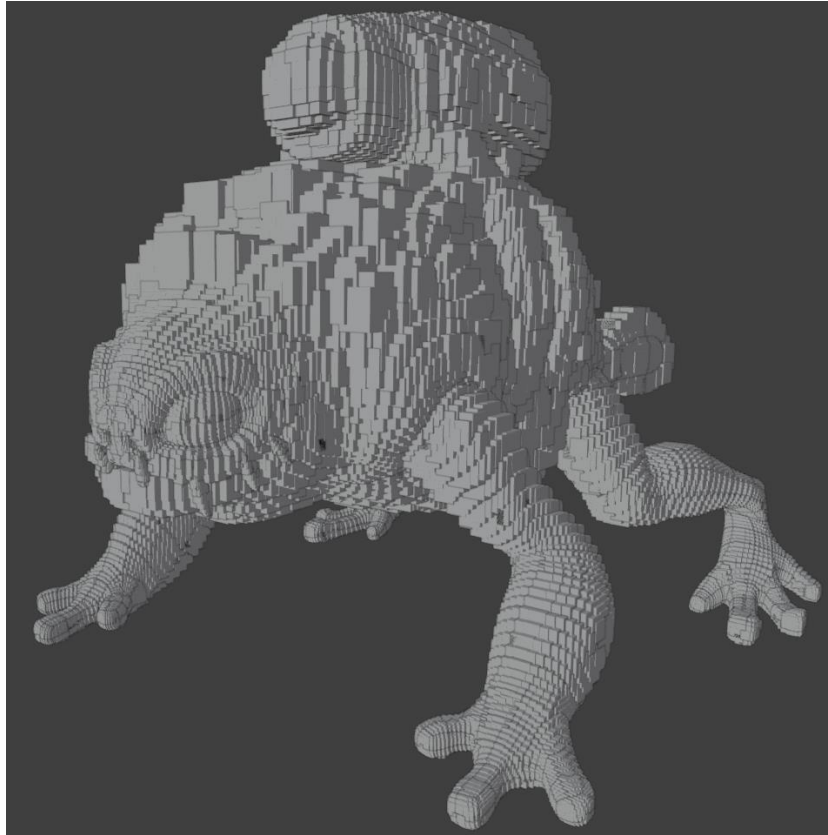
3 Compression Techniques

Box Approximation – Aligned Box Representation



3 Compression Techniques

Box Approximation – Aligned Box Representation



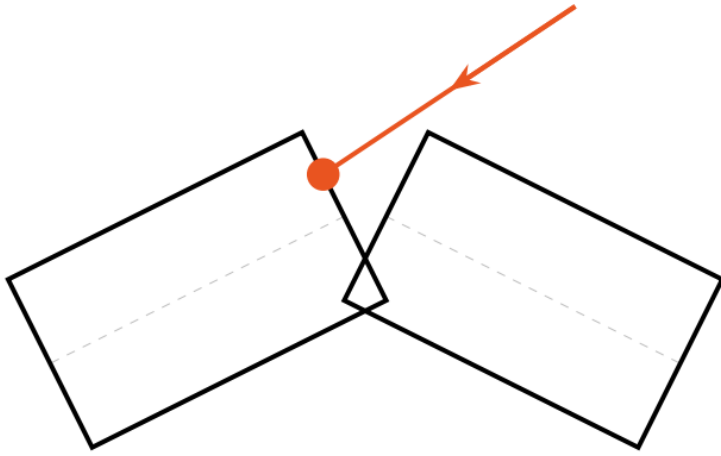
Basic Box Approximation



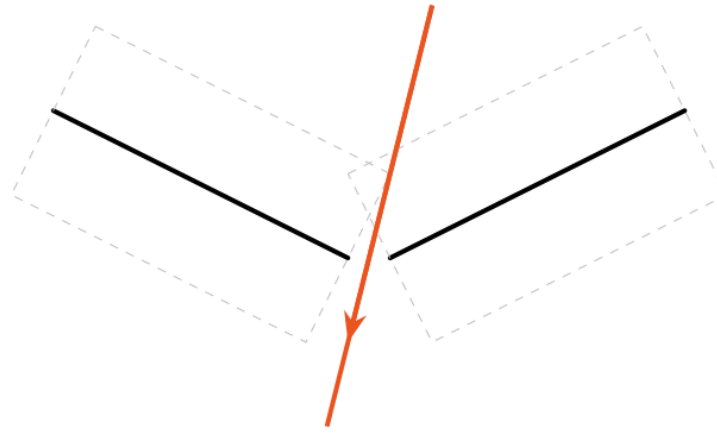
Approximation with
Aligned Bounding Boxes

3 Compression Techniques

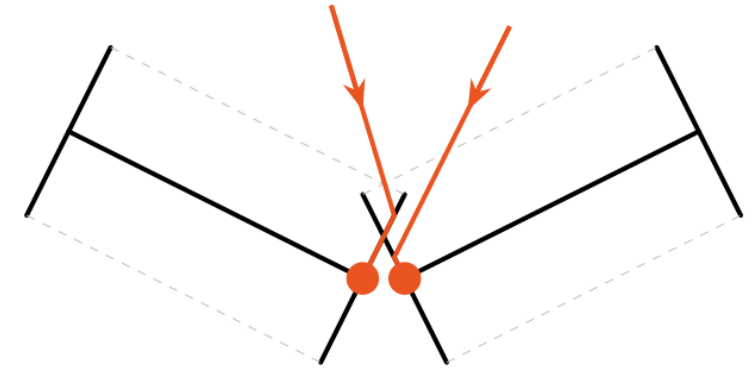
Box Approximation – Mid-Box Intersection



Basic Box
Approximation



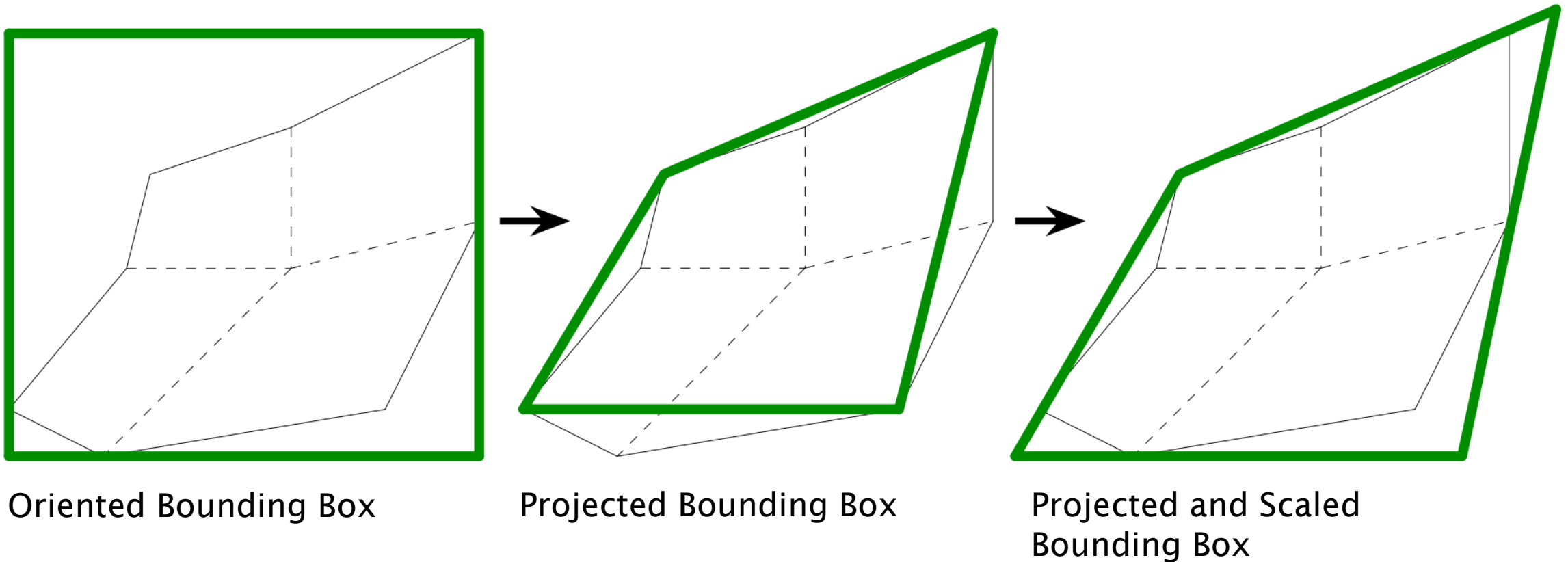
Problem of Cracks



Ray Projection to
Mid-Plane

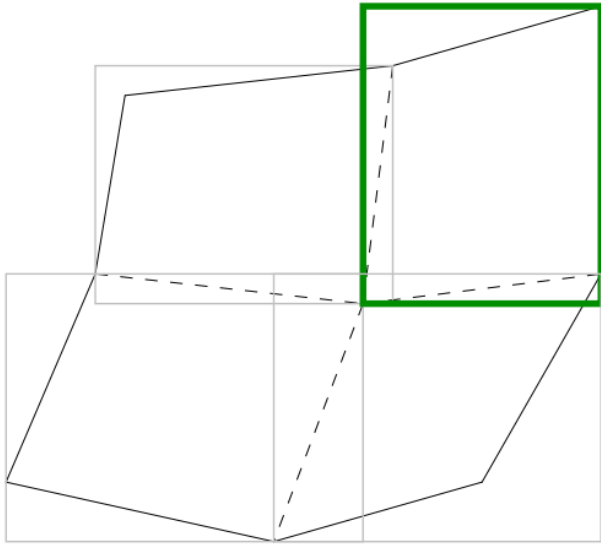
3 Compression Techniques

Box Approximation – Box Projection

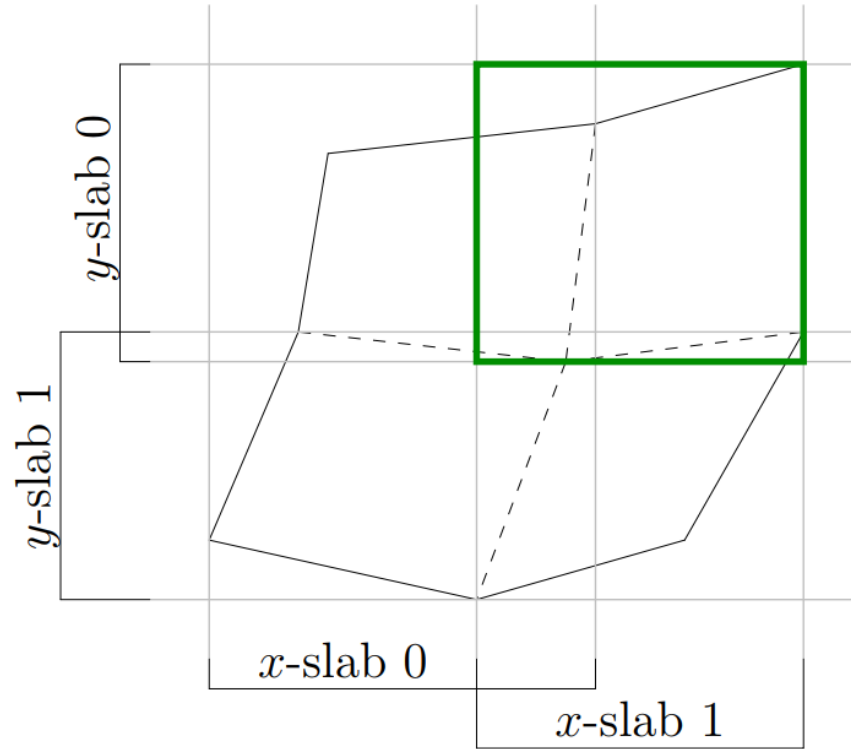


3 Compression Techniques

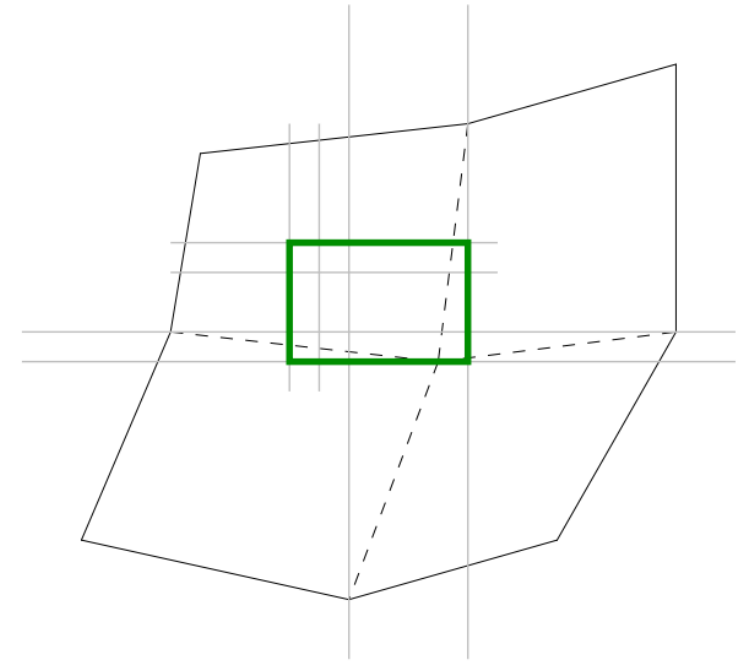
Compressed BVH – Slab Compression



Original Bounding Box
Node Size: 96 Byte



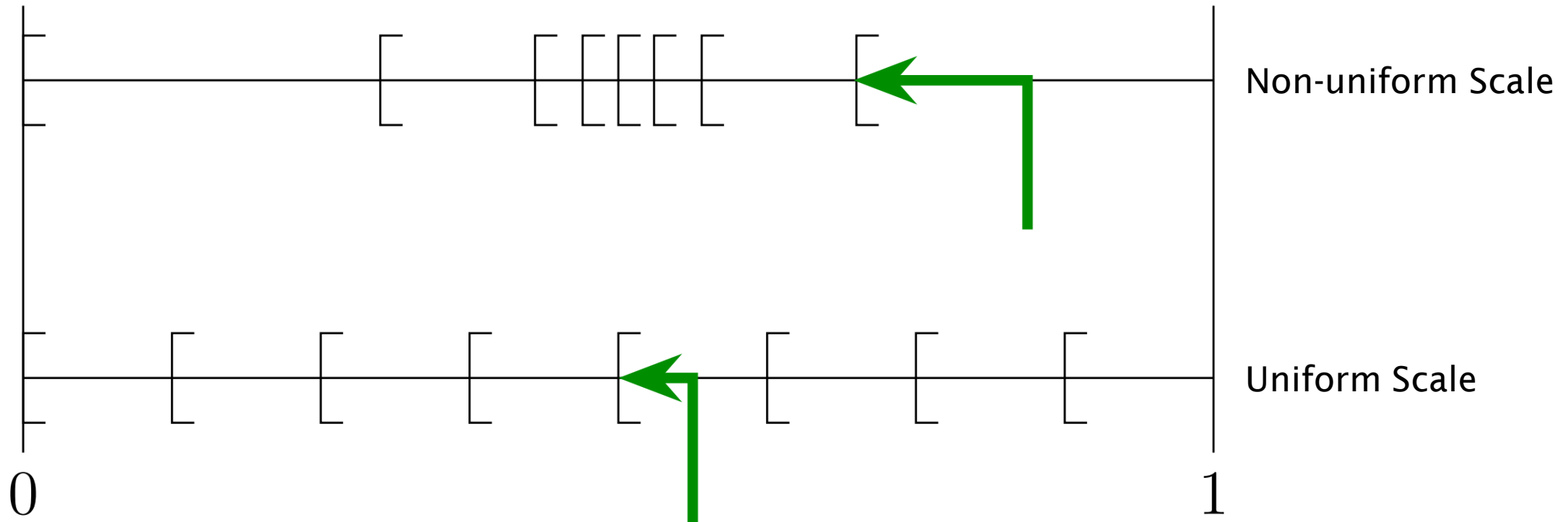
Full-Slab Compression
Node Size: 64 Byte



Half-Slab Compression
Node Size: 48 Byte

3 Compression Techniques

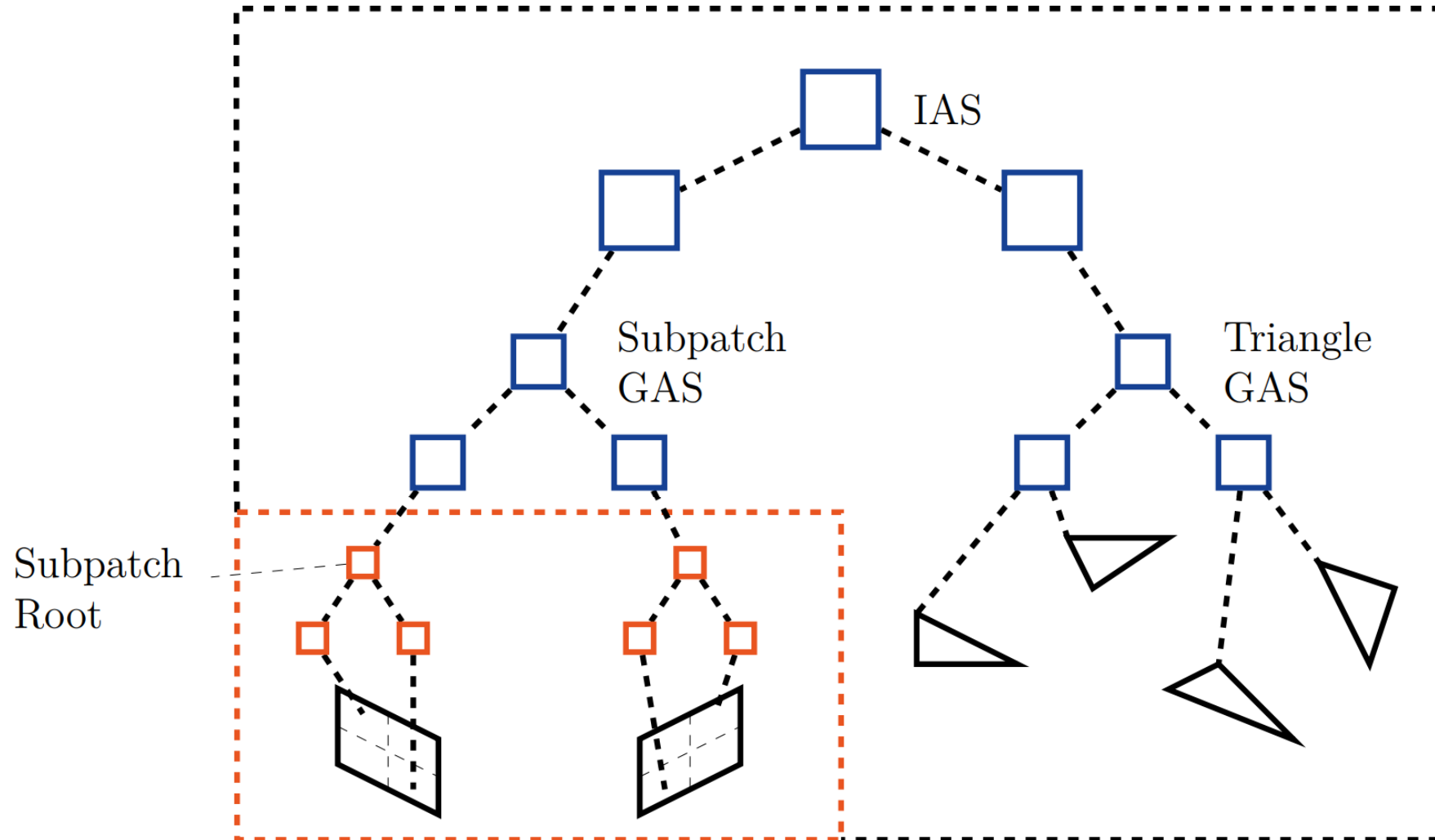
Compressed BVH - Quantization



Node Size: 2-4 Byte

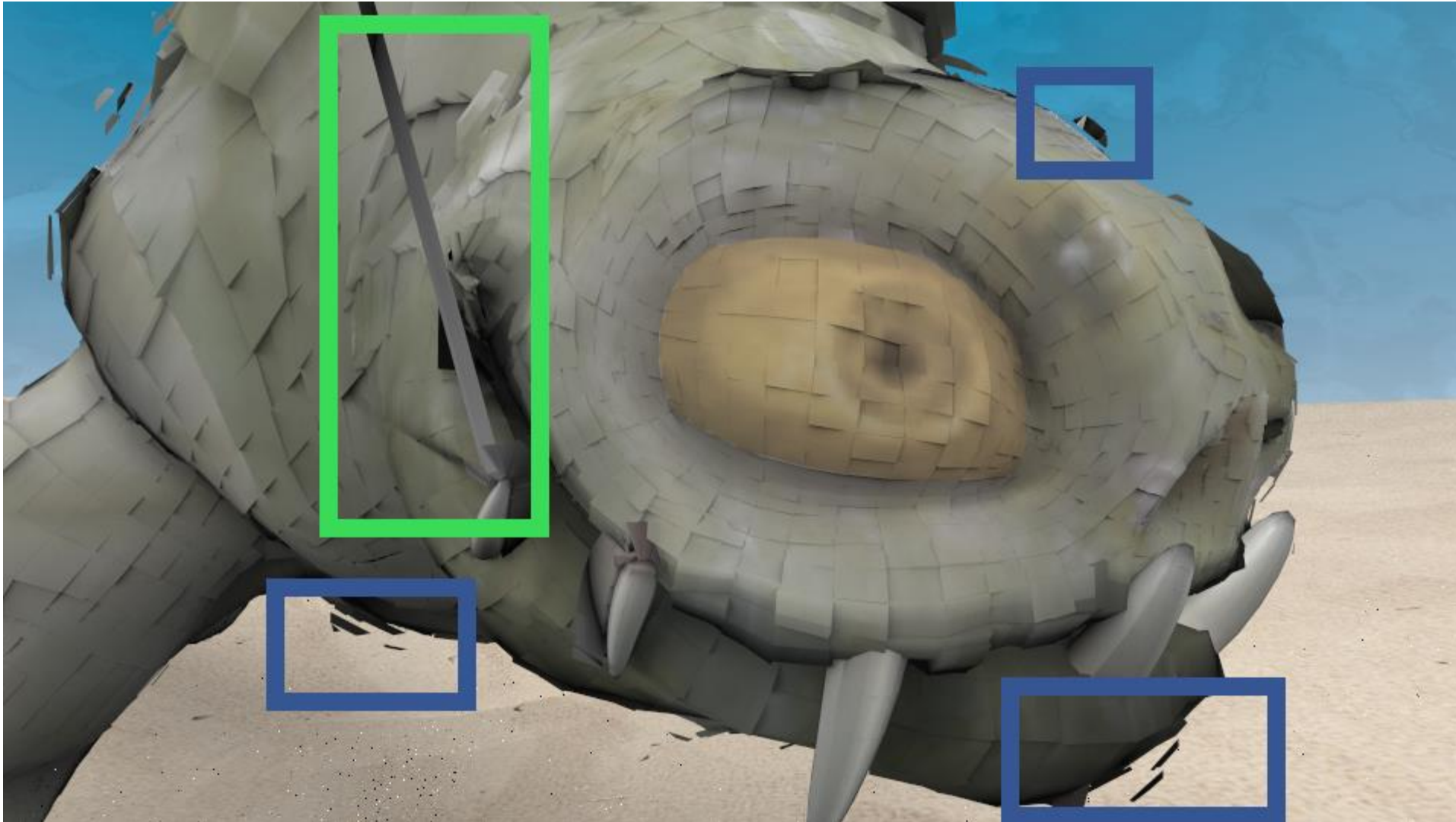
4 OptiX Implementation

Internal Acceleration Structure



5 Evaluation

Box Projection Artifacts



5 Evaluation

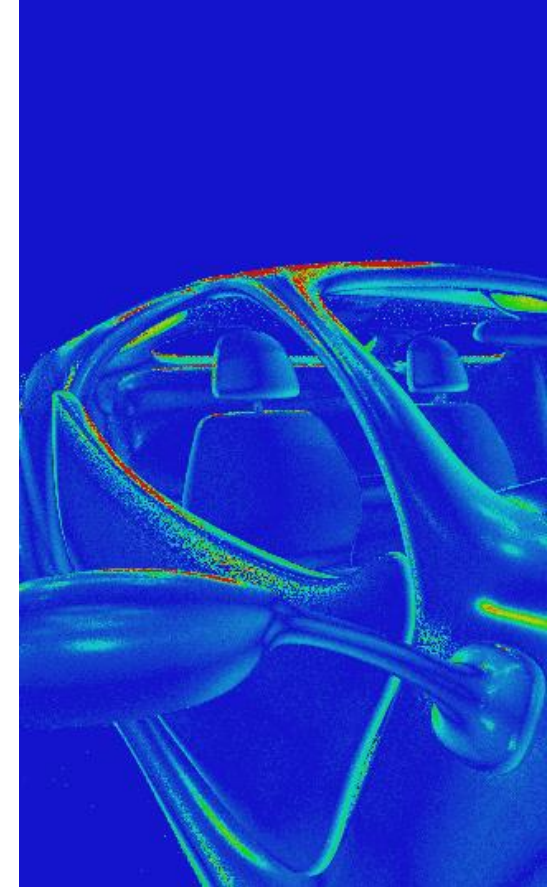
| Compression Scheme vs. Original Geometry



Original Geometry



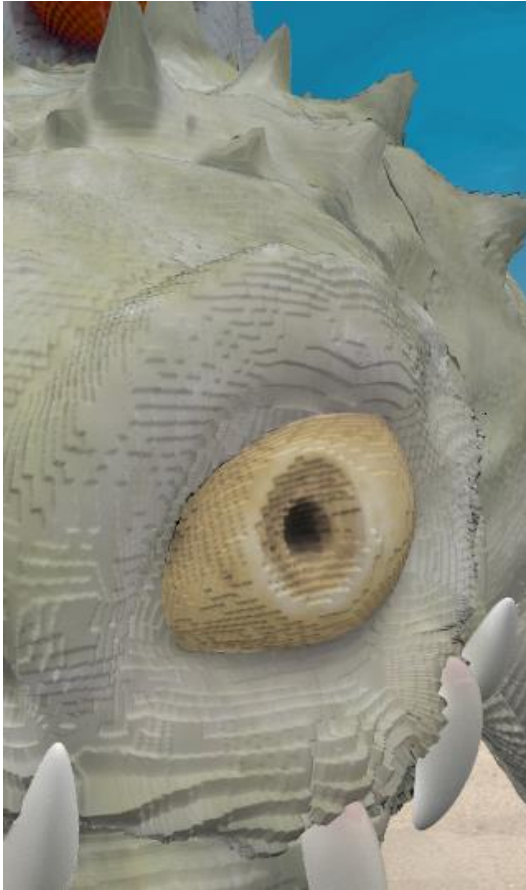
Compression Scheme



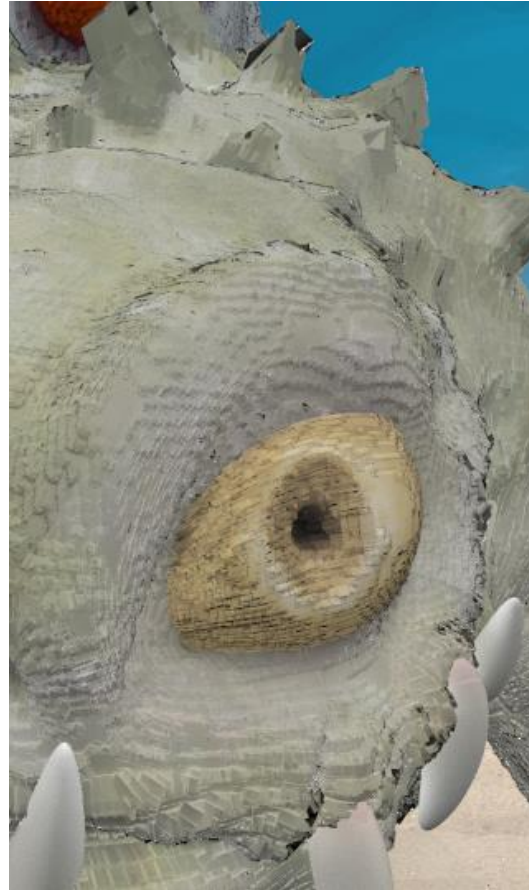
Difference

5 Evaluation

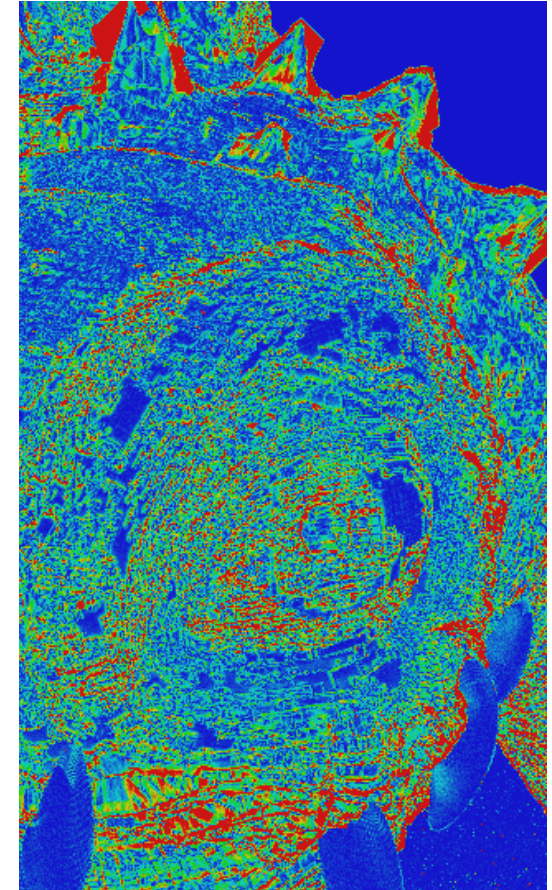
Compression Scheme vs. Original Geometry with Displacement



Original Geometry



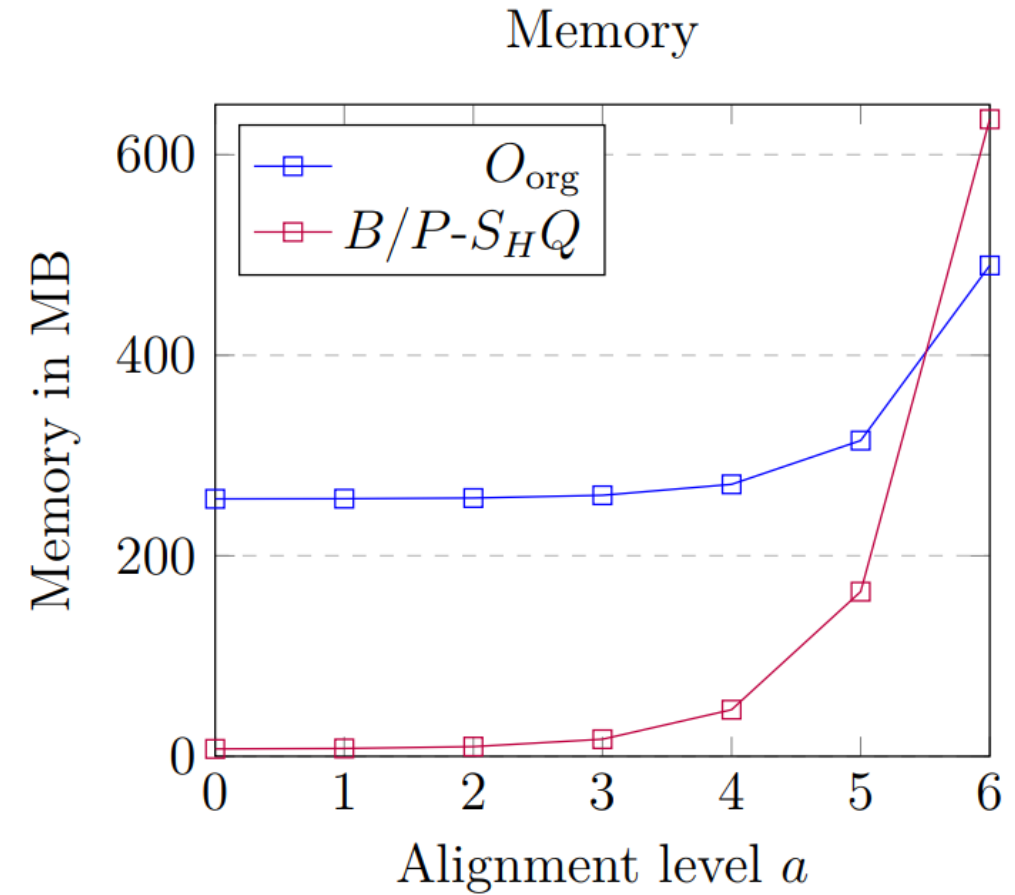
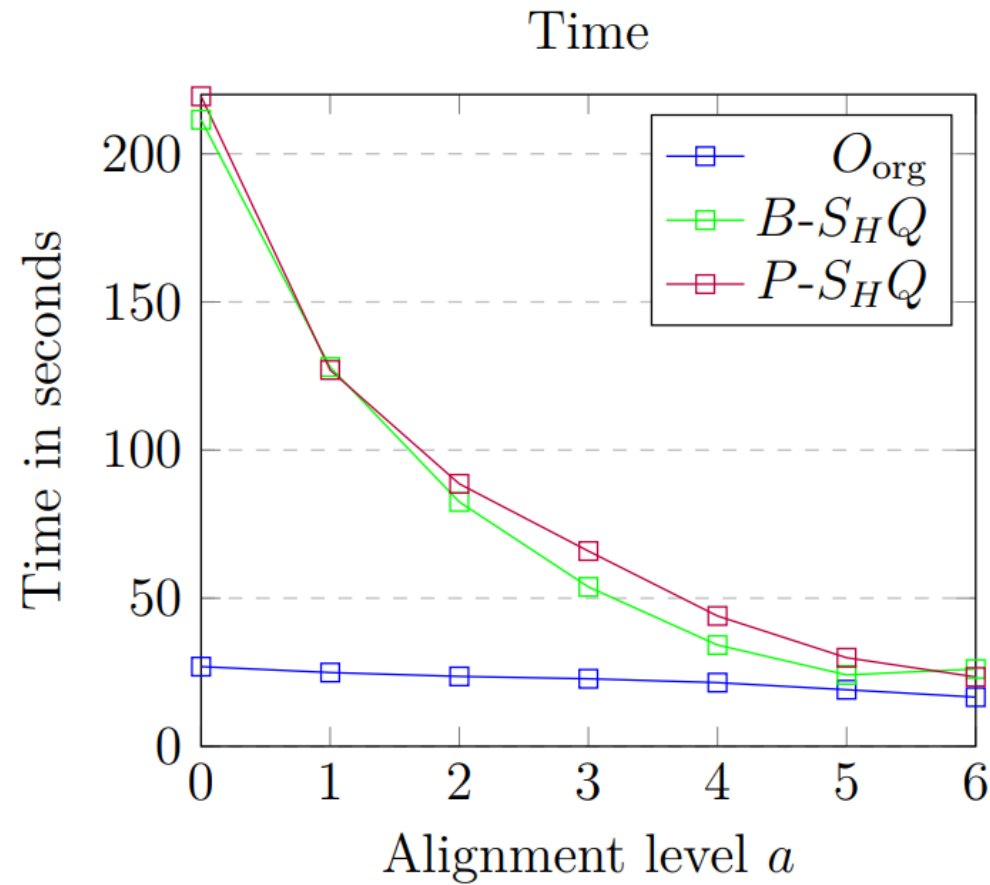
Compression Scheme
(without Box Projection)



Difference

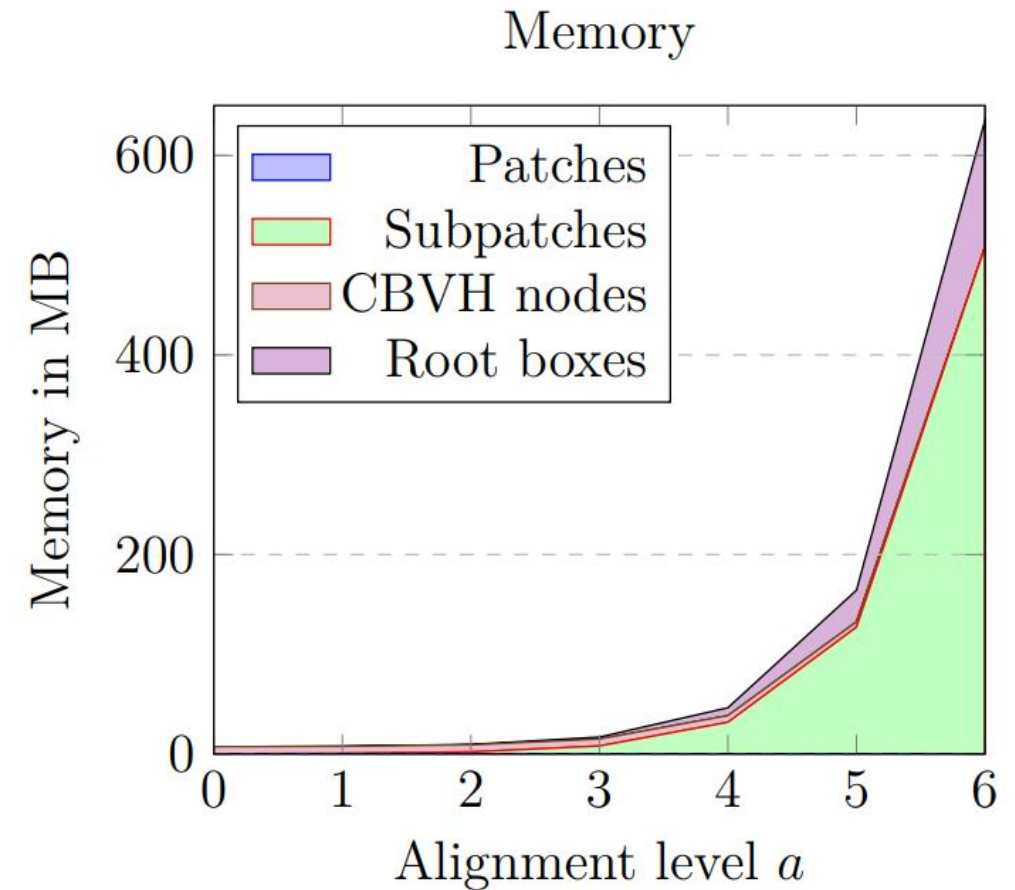
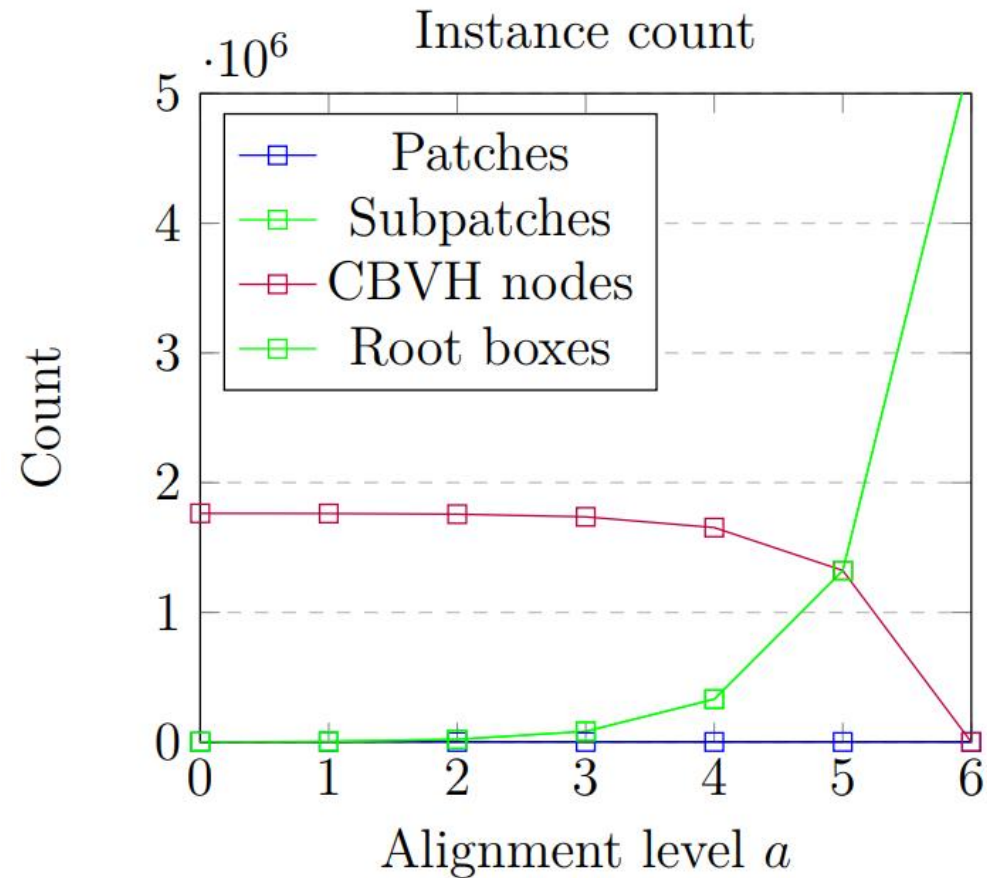
5 Evaluation

Temporal and Memory Comparison



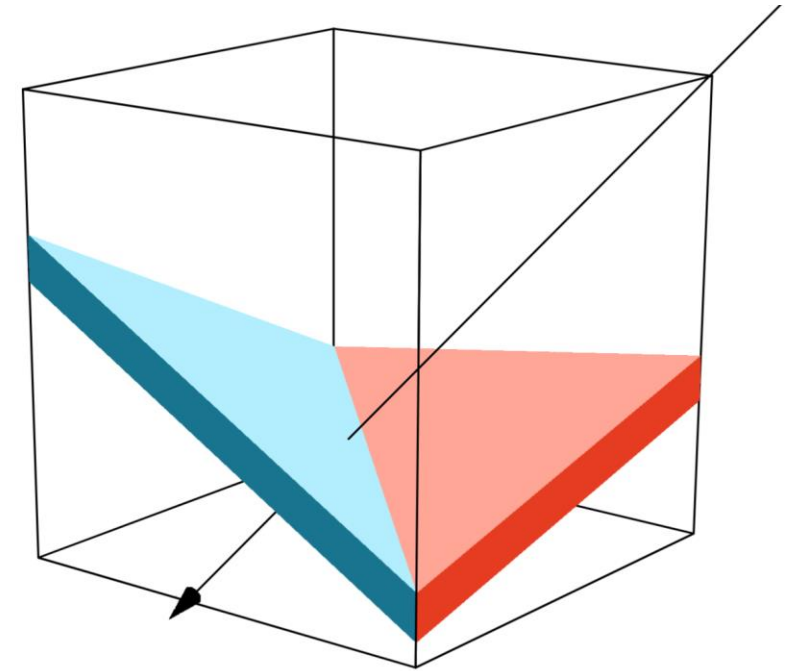
5 Evaluation

Resource Composition



6 Future Work [LMS*18]

| Refinement of the Compression Scheme for Displacement



Links

- Paper: A Compressed Representation for Ray Tracing Parametric Surfaces [SLM*16]
<https://doi.org/10.1145/2953877>
- Paper: A High-Resolution Compression Scheme for Ray Tracing Subdivision Surfaces with Displacement [LMS*18]
<https://doi.org/10.1145/3233308>
- NVIDIA OptiX Logo
<https://www.nvidia.com/de-de/gpu-cloud/hpc-visualization-containers/>